Hierarchical Geopolitical and Macro Multi-Asset Risk Engine

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0. Objective and scope

Objective: Build a systematic, event-triggered, multi-asset portfolio engine that (1) detects localized geopolitical, shipping/supply chain, sanctions, and macro regime shifts, (2) maps those shocks into tradable transmission channels, and (3) expresses the trade as a blended portfolio designed to maximize expected return subject to explicit risk, hedge, liquidity, and turnover constraints.

Core design constraint: The system must operate as "sleeves inside sleeves inside sleeves." It cannot be a single "shipping score" or "defense score." Example: shipping decomposes into chokepoints and port clusters (ex: India west-coast ports) with their own signal pipelines, their own models, their own instruments, and their own trade expressions.

Cadence: Signals update daily. Portfolio weights update weekly only when the event state machine indicates (a) impulse onset, (b) impulse decay, or (c) risk brakes force de-risking. Derivatives overlays can be activated intraweek only if an event impulse flips sharply and meets a high-confidence threshold.

Assets covered: ETFs, single names, government bond proxies, commodities proxies, crypto proxies, FX proxies where tradable, and derivatives overlays (options and futures where feasible, otherwise synthetic convexity via rules or leveraged/inverse products).

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1. High-level system architecture

The system is modular and hierarchical. Every module has a defined input contract, output contract, and evaluation standard so we can scale complexity without losing control.

Layer A: Data ingestion and caching

All raw data is ingested daily, transformed into consistent timestamps, and cached locally so backtests do not depend on live APIs after ingestion.

Daily inputs:

- Prices for all tradable instruments: close minimum, OHLCV preferred.
- Rates data: US yield proxies and curve metrics; ex-US government bond proxies where available.
- FX: USD proxy at minimum; optional EM FX baskets only if reliable and liquid.
- Macro risk: equity proxy, realized volatility, cross-asset correlation, cross-sectional dispersion.
- Event text stream: geopolitics, shipping/chokepoints, sanctions, central bank language, and trade restrictions.

Weekly feature aggregation:

- Robust statistics: median, trimmed mean, robust z-scores.
- Dynamics: week-over-week change, acceleration, novelty (new vs repeated coverage).
- Confirmations: “market already priced it” signals via vol/correlation/dispersion shifts.

Layer B: Hierarchical universe definition

We do not treat the instrument set as a flat universe. We define a hierarchy:

Level 0: Total portfolio
Level 1: Primary sleeves

- Defense & security
- Shipping & supply chain
- Commodities
- Crypto
- Rates (US)
- Rates (ex-US)
- Optional hedges (USD, equity beta, duration, tail-risk)

Level 2: Sub-sleeves (localized where possible)
Examples:

- Defense: primes, munitions/missile defense, ISR/space/comms, cyber/EW, naval/shipbuilding, nuclear enterprise modernization
- Shipping: chokepoints (Hormuz, Bab el-Mandeb, Suez, Black Sea, Taiwan Strait), regional routes (Indian Ocean, South China Sea), port clusters (India west coast, India east coast), sanctions enforcement, insurance/freight stress
- Commodities: energy complex, precious metals, industrial metals, agriculture only when event-implied
- Rates: duration points and country buckets
- Crypto: BTC core, ETH optional, plus “crypto regime” classification logic

Level 3: Instrument candidates per sub-sleeve
For each sub-sleeve we maintain:

- “Core ETF” candidates (broad exposure, high liquidity)
- “Satellite single-name” candidates (dispersion capture)
- “Derivative underlyings” (if options/futures are liquid enough)

This supports your preference: ETFs are the default base expression, single names are the alpha expression when catalysts are specific, and ETFs can also act as hedge expression when uncertainty is high.

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2. Exposure mapping (NLP-based, auditable, minimal manual work)

2.1 Why this exists

We need a mapping from instruments to themes and sub-themes without manually assigning exposure weights. The mapping must be stable enough for optimization, explainable enough to audit, and updateable without hand-maintaining hundreds of labels.

2.2 Text corpus per instrument

For each instrument we build a text bundle:

- Single names: 10-K business description + company overview text + reliable public summary
- ETFs: fund objective + holdings descriptions + sector tags + factsheet language

2.3 Theme ontology

Define a theme library with 40 to 120 themes depending on depth. This includes:

- Defense themes (munitions, missile defense, satellites, secure comms, cyber, shipbuilding, nuclear modernization infrastructure)
- Shipping themes (each chokepoint, each port cluster, insurance, freight, sanctions enforcement, rerouting)
- Macro/rates themes (hawkish shock, easing shock, curve inversion stress)
- Commodity themes (supply risk premium, safe-haven demand, industrial disruption)
- Crypto behavioral themes (risk-on beta vs capital flight behavior)

Each theme has:

- keyword set, anchor phrases, and disambiguation negatives
- a “theme description” paragraph used for semantic matching

2.4 Automated exposure estimation

For each instrument and each theme:

- Lexical match score (keyword and TF-IDF style similarity)
- Semantic score (embedding similarity between instrument text and theme description)
- Combine the two and normalize into an exposure vector per instrument

2.5 Audit rules (lightweight, team-based)

We do not manually fill maps. We do a controlled audit:

- Every rebalance cycle: review top weights and their top theme exposures
- Review exposures that changed the most since last refresh
- If wrong, update theme definitions and rerun mapping, not instrument-level overrides

This keeps the system systematic but not blind.

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3. Signal and event detection system (multi-expert, localized)

3.1 Core requirement: localized experts

Every meaningful sub-sleeve gets its own expert pipeline. Shipping does not borrow defense signals. Rates does not borrow shipping signals. The outputs are standardized, but inputs are not shared unless logically justified.

3.2 Expert outputs

Each expert produces a weekly output package:

- Activation probability: is this localized shock active next week
- Severity: expected magnitude of impact
- Confidence: reliability of the prediction this week
- Localization metadata: region, chokepoint, port cluster, event type

3.3 Example expert families

Conflict escalation experts

- Region-specific escalation intensity and acceleration
- Sanctions mentions and “tightening” vs “loosening” language
- Spillover flags (conflict plus energy corridor references)

Shipping and chokepoint experts

- Chokepoint-specific intensity and novelty signals
- Incident keyword clusters: attack, boarding, piracy, port closure, insurance, rerouting
- Port cluster stress: India west coast vs east coast treated as distinct nodes
- Market confirmation: energy moves, freight proxy moves, correlation changes

Rates and policy experts

- Yield changes, curve slope changes, inflation proxy drift
- Central bank language intensity proxies
- Regime classification: hawkish shock vs easing shock vs stable

Market pricing and complacency experts

- Realized volatility, vol of vol, cross-asset correlation spikes
- Dispersion across commodities, dispersion within defense single names
- “already priced” indicator used to reduce aggressiveness and push expression toward ETFs/hedges

Crypto regime experts

- Crypto volatility and correlation to equities
- USD moves and liquidity conditions
- Output: crypto behaving as risk-on vs behaving as stress hedge

3.4 Training labels for experts (no subjective “war week” labels)

Experts should be trained on measurable outcomes. Two practical approaches:

1. Define labels using cross-asset response patterns (example: shipping shock weeks defined by unusual combination of energy up, risk-off up, and shipping proxy abnormal returns).

2. Use unsupervised clustering on response vectors to discover regimes, then train experts to predict the regime cluster.

The key is consistency, not perfection. We care about actionable probability and direction, not storytelling.

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4. Stacked model architecture (experts feeding a combiner)

4.1 Why stacking is the right fit

You want multiple specialized experts, each capturing nonlinear interactions in their domain. You also want a stable, auditable layer that converts expert outputs into portfolio-level decisions. That is exactly what stacking does.

4.2 Base learners: random forests per expert

Each expert uses a random forest classifier/regressor as baseline because:

- nonlinear interactions matter
- threshold effects are common in event systems
- it tolerates messy feature distributions better than linear-only models

4.3 Meta-model: regularized linear combiner

The meta-model ingests expert outputs (probabilities, severity, confidence) and produces:

- theme-level attractiveness scores for next week, not direct weights
- optional "uncertainty" estimates used by the risk engine

Regularization is non-negotiable because expert outputs will be correlated. The combiner's job is stable weighting and arbitration.

4.4 Anti-leakage protocol

Stacking can look fake-good if you leak time. Our standard:

- rolling walk-forward training only
- expert predictions must be out-of-sample for the combiner training set
- no random shuffles
- no feature aggregation that peeks into the future

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5. Expression selection: ETF-first core + single-name satellites + ETF hedge, chosen by ML

This is the piece you specifically want deeper: how we decide ETF vs single names, and how ETFs act as a hedge if we might be wrong.

5.1 Expression modes per theme

For each theme each rebalance decision selects one of:

- ETF-only expression (core beta, low dispersion, high uncertainty)
- Single-name basket expression (high dispersion, specific catalyst, high confidence)
- Blend expression (single-name basket plus ETF hedge)

We can also allow: “single-name basket plus opposite-side hedge” if the theme naturally supports it (example: long defense subtheme, hedge broad market beta).

5.2 Model-driven expression selection

Expression selection is a classification model per theme with features like:

- expert confidence and severity
- dispersion inside candidate single names (are winners separating)
- liquidity and slippage proxies
- market stress and correlation regime

- earnings proximity penalty (single-name risk)
- “already priced” score from market pricing expert

Biasing toward single names: we bake the preference into the loss function or decision threshold so single names are chosen more often when dispersion and confidence are acceptable.

ETF as hedge concept: in blend mode, ETF is sized as an uncertainty hedge. Higher uncertainty increases the ETF hedge fraction and reduces single-name concentration.

5.3 Practical mechanics: building the single-name basket

For a theme, build candidates from:

- exposure map top-ranked names
- liquidity filters
- optional factor neutrality controls (keep market beta bounded if required)

Then allocate within basket using:

- exposure-weighted allocation
- or risk-parity inside the basket
- or model-predicted name-level scores if we extend the system to name-level predictions later

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6. Portfolio construction (hierarchical allocation + optimizer + risk engine)

6.1 What the optimizer actually does

The portfolio is built in stages, not in one flat pass.

Stage 1: Allocate risk budget across primary sleeves
Defense vs shipping vs commodities vs crypto vs rates vs hedges.

Stage 2: Allocate within sleeve to sub-sleeves
Example: shipping splits across chokepoints and port clusters; defense splits into munitions vs cyber vs naval, etc.

Stage 3: Expression choice within each sub-sleeve
ETF-only vs single names vs blend.

Stage 4: Final instrument weights
Apply constraints and finalize positions.

6.2 Constraints (explicit and enforceable)

- Fully invested or explicit cash allowance (choose one)
- Sleeve caps and sub-sleeve caps
- Instrument caps and dynamic caps tied to confidence
- Liquidity constraints and minimum volume thresholds
- Turnover constraints and cost penalties
- Hedge constraints: USD exposure targets, equity beta bounds, duration targets if desired
- Risk controls: target volatility, drawdown brakes, correlation limit to broad market if required

6.3 Dynamic caps driven by model confidence

You want fluid caps decided by the system. The safe version is:

- confidence expands allowable concentration
- low confidence forces broad ETF expressions and tighter caps
- hard global limits always remain, no exceptions

This gives “adaptive aggressiveness” without allowing the model to do something stupid.

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7. Derivatives layer: conditional convexity overlay

This is the new part. The goal is to “maximize” when we have a strong, localized view in a specific vertical, without making the whole system dependent on expensive options datasets.

7.1 Derivatives philosophy

Derivatives are not the main portfolio. They are a **convexity overlay** activated only when:

- the event impulse is fresh and localized
- expert confidence and severity clear a high threshold
- the expression selector prefers high convexity for that theme

7.2 What derivatives we use, realistically

There are three tiers:

Tier 1: Futures overlay (preferred for clean backtesting)

- Use futures or futures-like proxies to scale exposure efficiently when conviction is high.
- Applied mostly to rates and commodities where futures are standard.

Tier 2: Options overlay (only on very liquid underlyings)

- Options on major ETFs or the most liquid proxies.
- Use predefined structures with capped downside.

Tier 3: Synthetic convexity if historical options data is a bottleneck

- Vol targeting plus dynamic de-risk rules and event-driven leverage scaling.
- Leveraged/inverse ETFs for short holding windows only, with strict stop logic.

This structure is intentional: it keeps the strategy implementable and testable without paid data.

7.3 Options structures (simple, event-aligned)

For each activated theme, choose a structure based on directional confidence:

Direction clear

- Bullish: call spread or call with defined premium budget
- Bearish: put spread
  These cap premium and make costs predictable.

Direction unclear but magnitude high

- Strangle/straddle conceptually fits, but it is expensive and hard to backtest without IV history.
  We only allow this if we can justify it with a simplified cost model or we paper-trade it live going forward.

Already holding underlying

- Collar: sell call, buy put, reduce tail risk while maintaining exposure

7.4 Derivatives activation logic

Each theme gets a convexity score derived from:

- probability of activation
- severity
- confidence
- novelty (fresh impulse vs decaying hype)
- dispersion (single-name dispersion high suggests targeted expression)

If convexity score exceeds threshold:

- allocate a fixed premium budget (example: 10 to 50 bps of total portfolio per event cluster)
- enforce max total premium per week and max total derivatives exposure
- restrict holding period (event window) with an explicit unwind condition (impulse decay or time stop)

7.5 How derivatives integrate with ETF-first + single-name approach

A clean rule set:

- Base position: ETF core exposure for the theme
- Alpha layer: single-name basket when dispersion is high
- Hedge layer: ETF hedge fraction increases when uncertainty rises
- Convexity layer: options or futures overlay only when confidence and severity are extreme

This gives you a “beautiful” expression: core beta + targeted alpha + uncertainty hedge + convex kicker.

7.6 Backtesting derivatives without paid options history

We do two tracks:

- **Historical backtest:** backtest triggers and “convexity decisions” using underlying moves and conservative option cost assumptions (premium budget, spread assumptions, and capped payoff approximations).
- **Forward live paper-trade:** track real option prices going forward for a true PnL series and compare against the simplified historical approximation.

The key is to be honest in the doc about what is fully backtested and what is forward-tested.

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8. Event state machine (when we actually trade)

We do not rebalance every week by default. The model trades when the world state changes.

8.1 Impulse definition per expert

Each expert computes:

- intensity level
- acceleration
- novelty decay (repeated headlines count less)

8.2 Actions

- **Impulse onset:** activate new exposure, potentially activate derivatives overlay if convexity score triggers
- **Impulse continuation:** hold, only risk-scale and manage hedges
- **Impulse decay:** reduce exposure, unwind convexity overlay, rotate to neutral or alternate regime posture
- **Risk brake:** if realized vol or drawdown exceeds bounds, force de-risk regardless of event signals

This aligns with your preference: trades happen when events start or hype dies, not constant churn.

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9. Evaluation and "event replay" proof

9.1 Validation standards

- Walk-forward only with rolling retraining
- Costs and turnover enforced
- Liquidity filters enforced
- Out-of-sample performance is the only number that matters

9.2 Metrics

- Absolute return and Sharpe
- Max drawdown and recovery time
- Sleeve and sub-sleeve attribution
- Hedge overlay attribution (what hedges prevented)
- Derivatives overlay attribution (convexity benefit vs premium spent)
- Stability metrics: turnover, concentration, regime hit rate

9.3 Event replay library

For each major historical shock:

- localized expert signals leading into the event
- weights before and after event impulse
- which sleeves paid, which hedges paid, which derivatives overlays paid
- a clean narrative tied to measurable outputs, not vibes

This is the artifact that makes the project "knockout."

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10. Implementation responsibilities (parallelizable)

Expert modeling team

- build localized feature sets
- build labels and validation
- train and maintain RF experts
- standardize outputs

NLP exposure mapping team

- build theme ontology
- ingestion of instrument texts
- exposure vector generation and audit tooling
- quarterly refresh pipeline

Portfolio and risk team

- hierarchical allocator and optimizer

- dynamic caps and hedge constraints
- event state machine integration
- volatility targeting and drawdown brakes

Derivatives overlay team

- define eligible underlyings
- define overlay structures and sizing rules
- implement overlay activation logic
- historical approximation backtest and forward paper-trade tracker

Reporting and dashboard team

- signal dashboards (expert outputs per region and chokepoint)
- portfolio weights and changes
- attribution and event replay UI

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11. Summary of what makes this unique

- It is hierarchical and localized: chokepoints and ports are first-class objects, not generic scores.
- It is modular: many experts, standardized outputs, and a stable combiner.
- It is ETF-first but single-name biased, with ETF hedge logic built into the expression selector.
- It includes derivatives as a conditional convexity overlay, not as a gimmick.
- It is built to be proven by event replay: signal spike, portfolio shift, sleeve attribution, hedge attribution, convexity attribution.

Segmentation for Team

Part 1: Data backbone and universe setup (Anvay)

Difficulty: Medium-High

- Define the tradable universe and sleeve hierarchy
- Build an instrument registry that can return “all tickers for a sleeve or sub sleeve”
- Pull daily price data for all instruments
- Pull macro series like yields and USD proxy
- Cache everything locally so backtests never depend on live pulls
- Standardize timestamps, missing data handling, and file formats

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Part 2: Feature store and market state layer(Rayna)

Difficulty: Medium

- Convert daily data into weekly features
- Compute returns, volatility, vol of vol, drawdowns
- Compute cross asset correlation and dispersion metrics
- Build z scores, week over week changes, and acceleration features
- Output a clean weekly feature table used by every model
- Add basic feature validation checks so everyone is on the same inputs

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Part 3: Event and news signal pipeline (logan)

Difficulty: High

- Ingest event text streams and store them consistently
- Build keyword taxonomies for conflict, shipping, chokepoints, ports, sanctions, and rates language
- Generate localized event features by region, chokepoint, and port cluster
- Compute intensity, novelty, and acceleration of event coverage
- Produce a weekly event feature table that aligns with market weeks
- Create the impulse inputs that later drive the rebalance state machine

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Part 4: Expert model framework and expert suite (Joe)

Difficulty: High

- Implement the expert base class and the standard expert output format
- Build training and evaluation utilities shared across experts
- Ship the localized expert models
    - Conflict by region experts
    - Shipping chokepoint experts
    - Shipping port cluster experts
    - Sanctions and trade restriction experts
    - Rates regime experts
    - Market pricing and complacency experts
    - Commodity shock experts
    - Crypto regime experts
- Ensure every expert outputs the same core fields like probability active, severity, confidence, and tags

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Part 5: Meta model stacking, NLP exposure mapping, and expression selection

Difficulty: High

- Implement stacking so experts feed a meta model
- Make stacking leakage safe with rolling walk forward predictions
- Train a regularized combiner model that outputs theme scores
- Build NLP based exposure mapping from instrument text to themes
- Build audit tools to sanity check exposures for top weights
- Implement the ML expression selector that chooses ETF vs single names vs blend
- Add logic for ETF hedge fraction when uncertainty is high

Part 6: Portfolio engine, derivatives overlay, and backtest reporting

Difficulty: Very High

- Turn theme scores into instrument expected return scores
- Implement a constrained portfolio optimizer with caps, turnover limits, and risk targeting
- Add dynamic caps driven by confidence and regime
- Implement the rebalance state machine using event impulse onset and decay
- Add optional hedge sleeves like USD, duration, equity beta
- Implement derivatives overlay logic
    - activation rules for high confidence high severity events
    - structure selection like call spreads or put spreads or futures scaling
    - premium budget rules and unwind rules

- Build the full walk forward backtest harness
- Build attribution and event replay reports that show what signals fired, what trades happened, and what paid off